THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $3x + 4$	Linear means degree exactly 1.
2	**(a)** -10	Put x in $p(x) = 0$.
3	**(b)** -1	Substitute $x = -k$.
4	**(d)** 7	Put $x = 2$.
5	**(a)** $(4, 0)$	Put $y = 0$.
6	**(d)** 7	Find each zero.
7	**(d)** $6n - 5$	Use $a_n = a_1 + (n-1)d$.
8	**(b)** 250	Fare = fixed + rate × time.
9	**(c)** $5 - 2x$	Linear means degree exactly 1.
10	**(a)** -12	Put x in $p(x) = 0$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3x - 18 = 0$ gives $x = 6$ (1); The graph meets the x-axis at $(6, 0)$ (1).	1+1
12	$p(2) = 11$ (1); $p(-2) = -5$ (1).	1+1
13	Slope $= 8$, intercept $= 3$ (1); Cuts y-axis at $(0, 3)$ (1).	1+1
14	$5x + 20 = 0$ gives $x = -4$ (1); The graph meets the x-axis at $(-4, 0)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = -1$: $y = 1$; $x = 0$: $y = 3$ (1); $x = 1$: $y = 5$; $x = 2$: $y = 7$ (1); Slope $= 2$, y-intercept $= 3$ (1).	1+1+1
16	Substitute: $11 = 6 + k$ (1); $k = 5$ (1); Line: $y = 2x + 5$ (1).	1+1+1
17	$p + q = 5x + 10$ (1); Put $5x + 10 = 0$ (1); Zero $x = -2$ (1).	1+1+1
	OR: $x = -1$: $y = -4$; $x = 0$: $y = -1$ (1); $x = 1$: $y = 2$; $x = 2$: $y = 5$ (1); Slope $= 3$, y-intercept $= -1$ (1).	1+1+1
18	Substitute: $4 = 6 + k$ (1); $k = -2$ (1); Line: $y = 3x - 2$ (1).	1+1+1
	OR: $p + q = 4x - 4$ (1); Put $4x - 4 = 0$ (1); Zero $x = 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$C(x) = 12x + 60$ (1); $C(5) = 120$ (1); $12x + 60 = 168$ (1); $x = 9$ (1); Hence the number of notebooks is 9 (1).	1+1+1+1+1
	OR: $y(0) = -12$, $y(1) = -8$, $y(2) = -4$ (1); Crosses x-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 4$ (1); Slope is positive: the graph rises (1).	1+1+1+1+1
20	$C(x) = 15x + 45$ (1); $C(4) = 105$ (1); $15x + 45 = 150$ (1); $x = 7$ (1); Hence the number of notebooks is 7 (1).	1+1+1+1+1
	OR: $y(0) = 9$, $y(1) = 6$, $y(2) = 3$ (1); Crosses x-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= -3$ (1); Slope is negative: the graph falls (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -1	Substitute $x = -k$.
2	**(c)** 6	Put $x = 2$.
3	**(b)** $(-4, 0)$	Put $y = 0$.
4	**(d)** 8	Find each zero.
5	**(d)** $3n + 5$	Use $a_n = a_1 + (n-1)d$.
6	**(b)** 390	Fare = fixed + rate × time.
7	**(b)** $\frac{x}{3} + 1$	Linear means degree exactly 1.
8	**(a)** -10	Put x in $p(x) = 0$.
9	**(c)** -14	Substitute $x = -k$.
10	**(c)** 13	Put $x = 2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(4) = 7$ (1); $p(-4) = -17$ (1).	1+1
12	Slope $= -5$, intercept $= 9$ (1); Cuts y-axis at $(0, 9)$ (1).	1+1
13	$2x + 14 = 0$ gives $x = -7$ (1); The graph meets the x-axis at $(-7, 0)$ (1).	1+1
14	$p(1) = 8$ (1); $p(-1) = -4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = -1$: $y = 7$; $x = 0$: $y = 5$ (1); $x = 1$: $y = 3$; $x = 2$: $y = 1$ (1); Slope $= -2$, y-intercept $= 5$ (1).	1+1+1
16	Substitute: $9 = 4 + k$ (1); $k = 5$ (1); Line: $y = 4x + 5$ (1).	1+1+1
17	$p + q = 6x + 6$ (1); Put $6x + 6 = 0$ (1); Zero $x = -1$ (1).	1+1+1
	OR: $x = -1$: $y = -2$; $x = 0$: $y = 2$ (1); $x = 1$: $y = 6$; $x = 2$: $y = 10$ (1); Slope $= 4$, y-intercept $= 2$ (1).	1+1+1
18	Substitute: $6 = 15 + k$ (1); $k = -9$ (1); Line: $y = 5x - 9$ (1).	1+1+1
	OR: $p + q = 6x + 6$ (1); Put $6x + 6 = 0$ (1); Zero $x = -1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$C(x) = 20x + 90$ (1); $C(3) = 150$ (1); $20x + 90 = 250$ (1); $x = 8$ (1); Hence the number of notebooks is 8 (1).	1+1+1+1+1
	OR: $y(0) = 10$, $y(1) = 15$, $y(2) = 20$ (1); Crosses x-axis: $x = -2$ (1); Point $(-2, 0)$ (1); Slope $= 5$ (1); Slope is positive: the graph rises (1).	1+1+1+1+1
20	$C(x) = 14x + 70$ (1); $C(6) = 154$ (1); $14x + 70 = 210$ (1); $x = 10$ (1); Hence the number of notebooks is 10 (1).	1+1+1+1+1
	OR: $y(0) = -8$, $y(1) = -10$, $y(2) = -12$ (1); Crosses x-axis: $x = -4$ (1); Point $(-4, 0)$ (1); Slope $= -2$ (1); Slope is negative: the graph falls (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $(5, 0)$	Put $y = 0$.
2	**(b)** 10	Find each zero.
3	**(a)** $9n - 6$	Use $a_n = a_1 + (n-1)d$.
4	**(b)** 280	Fare = fixed + rate × time.
5	**(b)** $4x$	Linear means degree exactly 1.
6	**(b)** -24	Put x in $p(x) = 0$.
7	**(a)** -11	Substitute $x = -k$.
8	**(c)** 5	Put $x = 2$.
9	**(d)** $(-4, 0)$	Put $y = 0$.
10	**(b)** 13	Find each zero.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slope $= 9$, intercept $= -2$ (1); Cuts y-axis at $(0, -2)$ (1).	1+1
12	$7x - 28 = 0$ gives $x = 4$ (1); The graph meets the x-axis at $(4, 0)$ (1).	1+1
13	$p(3) = 22$ (1); $p(-3) = -8$ (1).	1+1
14	Slope $= -7$, intercept $= -4$ (1); Cuts y-axis at $(0, -4)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = -1$: $y = -8$; $x = 0$: $y = -3$ (1); $x = 1$: $y = 2$; $x = 2$: $y = 7$ (1); Slope $= 5$, y-intercept $= -3$ (1).	1+1+1
16	Substitute: $3 = 10 + k$ (1); $k = -7$ (1); Line: $y = 2x - 7$ (1).	1+1+1
17	$p + q = 6x - 6$ (1); Put $6x - 6 = 0$ (1); Zero $x = 1$ (1).	1+1+1
	OR: $x = -1$: $y = 9$; $x = 0$: $y = 6$ (1); $x = 1$: $y = 3$; $x = 2$: $y = 0$ (1); Slope $= -3$, y-intercept $= 6$ (1).	1+1+1
18	Substitute: $20 = 12 + k$ (1); $k = 8$ (1); Line: $y = 3x + 8$ (1).	1+1+1
	OR: $p + q = 5x - 10$ (1); Put $5x - 10 = 0$ (1); Zero $x = 2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$C(x) = 11x + 55$ (1); $C(7) = 132$ (1); $11x + 55 = 121$ (1); $x = 6$ (1); Hence the number of notebooks is 6 (1).	1+1+1+1+1
	OR: $y(0) = -18$, $y(1) = -12$, $y(2) = -6$ (1); Crosses x-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 6$ (1); Slope is positive: the graph rises (1).	1+1+1+1+1
20	$C(x) = 25x + 120$ (1); $C(2) = 170$ (1); $25x + 120 = 395$ (1); $x = 11$ (1); Hence the number of notebooks is 11 (1).	1+1+1+1+1
	OR: $y(0) = 20$, $y(1) = 16$, $y(2) = 12$ (1); Crosses x-axis: $x = 5$ (1); Point $(5, 0)$ (1); Slope $= -4$ (1); Slope is negative: the graph falls (1).	1+1+1+1+1

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